1 Hyperparameter selection

As mentioned in the main paper, we perform hyperparameter selection for all methods through grid search combined with a hold-out strategy. Specifically, we randomly select 20% of the training data as validation set and use the remaining 80% for training. The hyperparameter configuration that achieves the best Youden's J statistic on the validation set is selected for each method. The hyperparameter grids considered for each method are as follows:

The tested parameters for **MCTS4SD** and their respective values are: *Redundancy Strategy* (jaccardSupportDescription, jaccardSupportDescriptionTarget, sumJaccard), *Max Redundancy* (0.4, 0.6, 0.8), *Number of iterations* (1000, 10000, 100000), and *Measure* (WRAcc, WeightedRelativeF1, RelativeF1, Acc). The other parameters were kept at their default values as in the implementation provided by [**?**].

For **DiffNaps**, we tested the following parameters: *Hidden dimensions* (500, 1000, 2000, 5000, 10000), *Number of epochs* (25, 50, 75, 100), *Weight decay* (5.0, 8.0, 10.0, 12.0), *Learning rate* (0.001, 0.005, 0.01), and *Batch size* (32, 64, 128). The rest of the parameters were kept at their default values as in the implementation provided by [**?**]

The hyperparameters for the **Beam search** method are: *Beam width* (10, 25, 100, 250), *Measure* (chisquaredQF, standardQF ($a = 0.5$), standardQF ($a = 1$)), and *Number of subgroups* (3, 5, 10, 20, 50). The maximum subgroup size was set to 3 for all datasets.

For **SDIGA**, we tested the following parameters: *Confidence weight* (0.3, 0.4, 0.5), and *Support weight* (0.3, 0.4, 0.5). For performance reasons, we set the following parameters to fixed values: population size = 7000, number of generations = 500, maximum subgroup size = 10, and maximum number of returned subgroups = 30. Crossover and mutation rates were set to 0.9 and 0.1, respectively.

Regarding our proposed method **NeuralSD**, we only evaluated the *Quality Measure* (WRAcc, Qc ($c = 0.5, 0.9, 2.0, 5.0, 10.0$)). The number of returned subgroup size was set to 10 for all datasets. In each iteration, a minimum quality of 10 (Qc) or 0.01 ($WRAcc$) was required to consider a subgroup as valid and add it to the result set, otherwise the search process is stopped. The learning rate was set to 100 and, while the maximum number of epochs per iteration was set to 1000, early stopping with a patience of 2 epochs was also applied to speed up the training process. Regarding the Facebook dataset, and given its higher number of instances, we set a batch size of 1024 for training NeuralSD to speed up the process.

The final hyperparameter configurations selected for each dataset and algorithm are reported in Tables 1, 2, 3, and 4.

Table 1: Hyperparameters selected for Facebook dataset

Parameter	MCTS	DiffNaps	Beam	SDIGA	NSD
Quality measure	WRAcc	-	WRAcc	-	WRAcc
Number of iterations	1000	-	-	-	-
Redundancy strategy	jaccard Support Description	-	-	-	-
Maximum redundancy	0.4	-	-	-	-
Hidden dim.	-	2000	-	-	-
Epochs	-	100	-	-	-
Weight decay	-	10	-	-	-
Learning rate	-	0.001	-	-	-
Batch size	-	32	-	-	-
Beam width	-	-	100	-	-
Num. subroups	-	-	3	-	-
Conf weight	-	-	-	0.3	-
Supp weight	-	-	-	0.5	-

Table 2: Hyperparameters selected for Synthetic dataset

Parameter	MCTS	DiffNaps	Beam	SDIGA	NSD
Quality measure	WRAcc	-	standQF a = 0.5	-	Qc c = 0.9
Number of iterations	100000	-	-	-	-
Redundancy strategy	jaccard Support Description	-	-	-	-
Maximum redundancy	0.4	-	-	-	-
Hidden dim.	-	10000	-	-	-
Epochs	-	75	-	-	-
Weight decay	-	10	-	-	-
Learning rate	-	0.005	-	-	-
Batch size	-	64	-	-	-
Beam width	-	-	250	-	-
Num. subroups	-	-	50	-	-
Conf weight	-	-	-	0.5	-
Supp weight	-	-	-	0.4	-

Table 3: Hyperparameters selected for Noisy Synthetic dataset

Parameter	MCTS	DiffNaps	Beam	SDIGA	NSD
Quality measure	WRAcc	-	standQF a = 0.5	-	Qc c = 0.9
Number of iterations	100000	-	-	-	-
Redundancy strategy	sum Jaccard	-	-	-	-
Maximum redundancy	0.4	-	-	-	-
Hidden dim.	-	500	-	-	-
Epochs	-	50	-	-	-
Weight decay	-	5	-	-	-
Learning rate	-	0.01	-	-	-
Batch size	-	128	-	-	-
Beam width	-	-	100	-	-
Num. subroups	-	-	20	-	-
Conf weight	-	-	-	0.3	-
Supp weight	-	-	-	0.4	-

Table 4: Hyperparameters selected for BRCA_N dataset

Parameter	MCTS	DiffNaps	Beam	SDIGA	NSD
Quality measure	WRAcc	-	standQF a = 0.5	-	Qc c = 0.9
Number of iterations	1000	-	-	-	-
Redundancy strategy	jaccard Support Description	-	-	-	-
Maximum redundancy	0.4	-	-	-	-
Hidden dim.	-	10000	-	-	-
Epochs	-	100	-	-	-
Weight decay	-	10	-	-	-
Learning rate	-	0.005	-	-	-
Batch size	-	64	-	-	-
Beam width	-	-	25	-	-
Num. subroups	-	-	10	-	-
Conf weight	-	-	-	0.5	-
Supp weight	-	-	-	0.4	-